Word2vec learns dense embeddings from context

Running text supplies the labels; the learned classifier weights become word vectors.

Sparse count vector

`[0, 0, 442, 0, 8, ...]`

$|V|$ dimensions · mostly zeroseach dimension names a context word

learn rather than count

Dense word2vec embedding

`[0.31, -0.08, 0.74, ...]`

$d \approx 50\text{–}1000$ · dense real valueslearned dimensions need not be interpretable

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Skip-gram with negative sampling (SGNS)

Running text creates the labels: self-supervision

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Skip-gram with negative sampling (SGNS)

Running text creates the labels: self-supervision

1 Positive pairs

Nearby words
(apricot, jam) → +

Word2vec learns dense embeddings from context

Running text supplies the labels; the learned classifier weights become word vectors.

Sparse count vector

`[0, 0, 442, 0, 8, ...]`

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Compact vectors reduce parameters and may capture synonymy better.

Skip-gram with negative sampling (SGNS)

Running text creates the labels: self-supervision

1 Positive pairs

Nearby words (apricot, jam) → +

→

2 Negative pairs

Random samples (apricot, aardvark) → -

→

Word2vec learns dense embeddings from context

Running text supplies the labels; the learned classifier weights become word vectors.

Sparse count vector

`[0, 0, 442, 0, 8, ...]`

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Skip-gram with negative sampling (SGNS)

Running text creates the labels: self-supervision

1 Positive pairs

Nearby words (apricot, jam) → +

→

2 Negative pairs

Random samples (apricot, aardvark) → -

→

3 Binary classifier

Logistic regression separates + from -

→

Word2vec learns dense embeddings from context

Running text supplies the labels; the learned classifier weights become word vectors.

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Compact vectors reduce parameters and may capture synonymy better.

Skip-gram with negative sampling (SGNS)

Running text creates the labels: self-supervision

1 Positive pairs

Nearby words (apricot, jam) → +

→

2 Negative pairs

Random samples (apricot, aardvark) → -

→

3 Binary classifier

Logistic regression separates + from -

→

4 Keep the weights

One fixed vector per word (static embedding)

Skip-gram asks whether two words belong together

A target word and a candidate context word define one binary classification example.

A ±2 context window supplies observed pairs

…	lemon,	a	tablespoon	of	apricot	jam,	a	pinch	…
			c_1	c_2	target w	c_3	c_4		

Skip-gram asks whether two words belong together

A target word and a candidate context word define one binary classification example.

A ±2 context window supplies observed pairs

…	lemon,	a	tablespoon	of	apricot	jam,	a	pinch	…
			c_1	c_2	target w	c_3	c_4		

Observed nearby

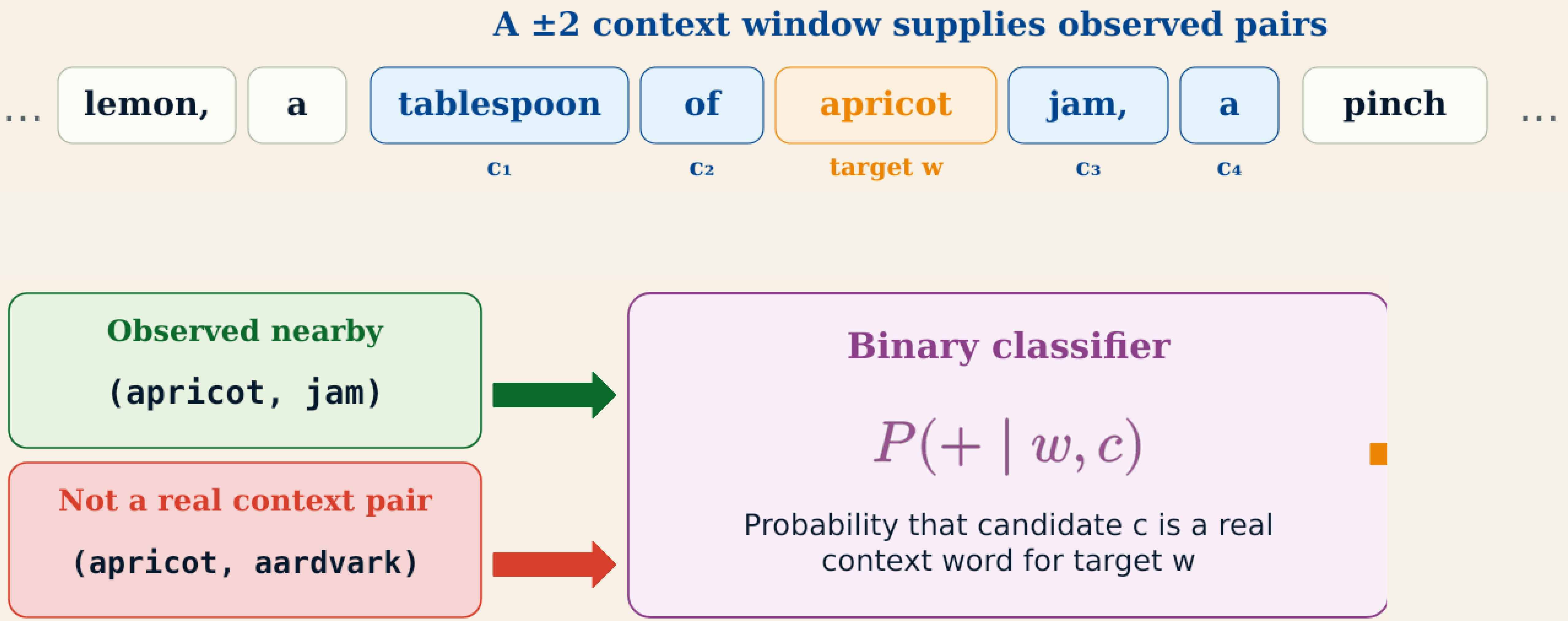
`(apricot, jam)`

Not a real context pair

`(apricot, aardvark)`

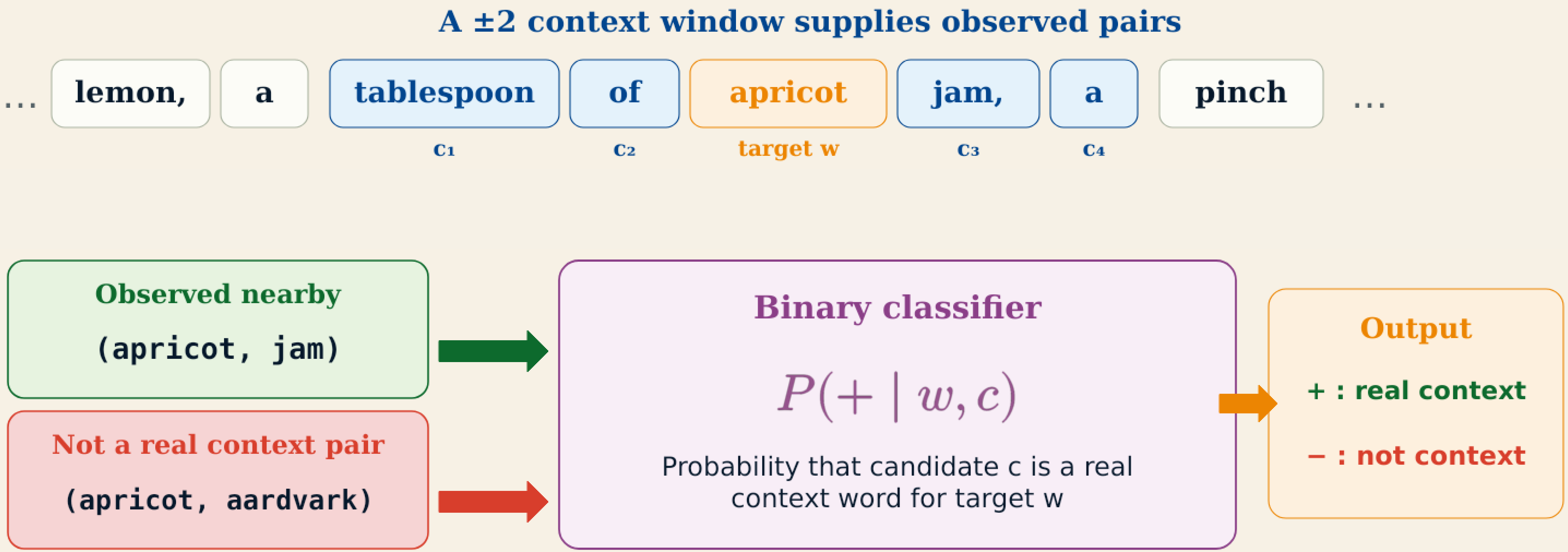
Skip-gram asks whether two words belong together

A target word and a candidate context word define one binary classification example.



Skip-gram asks whether two words belong together

A target word and a candidate context word define one binary classification example.



Skip-gram asks whether two words belong together

A target word and a candidate context word define one binary classification example.

A ±2 context window supplies observed pairs

… lemon, a **tablespoon** **of** **apricot** **jam,** **a** pinch …

c_1 (tablespoon) · c_2 (of) · target w (apricot) · c_3 (jam,) · c_4 (a)

Observed nearby

`(apricot, jam)`

Not a real context pair

`(apricot, aardvark)`

Binary classifier

$$P(+ \mid w, c)$$

Probability that candidate c is a real context word for target w

Output

+ : real context

− : not context

J&M §5.5.1

Where do the vectors *w* and *c* used by this classifier come from?

Every word has a target and a context embedding

Skip-gram learns separate parameter matrices W and C for these two roles.

$\theta =$

W target	dimensions 1 … d
aardvark	· · … ·
apricot	0.31 −0.08 … 0.74
…	…
zebra	· · … ·

Every word has a target and a context embedding

Skip-gram learns separate parameter matrices W and C for these two roles.

$\theta =$

W target	dimensions 1 … d			
aardvark	·	·	…	·
apricot	0.31	−0.08	…	0.74
…	…			
zebra	·	·	…	·

Every word has a target and a context embedding

Skip-gram learns separate parameter matrices W and C for these two roles.

$\theta =$

W target	dimensions 1 … d
aardvark	· · … ·
apricot	0.31 −0.08 … 0.74
…	…
zebra	· · … ·

Target lookup in W

$$\mathbf{w} = W[\text{apricot}]$$

apricot used as the target

Every word has a target and a context embedding

Skip-gram learns separate parameter matrices W and C for these two roles.

$\theta =$

W target	dimensions 1 … d
aardvark	· · … ·
apricot	0.31 −0.08 … 0.74
…	…
zebra	· · … ·

C context	dimensions 1 … d
aardvark	· · … ·
jam	−0.12 0.57 … 0.23
…	…
zebra	· · … ·

Target lookup in W

$$\mathbf{w} = W[\text{apricot}]$$

apricot used as the target

Every word has a target and a context embedding

Skip-gram learns separate parameter matrices W and C for these two roles.

$$\theta =$$

W target	**dimensions 1 … d**
aardvark	· · … ·
apricot	0.31 −0.08 … 0.74
…	…
zebra	· · … ·

C context	**dimensions 1 … d**
aardvark	· · … ·
jam	−0.12 0.57 … 0.23
…	…
zebra	· · … ·

Target lookup in W

$$\mathbf{w} = W[\text{apricot}]$$

apricot used as the target

Context lookup in C

$$\mathbf{c} = C[\text{jam}]$$

jam used as context or noise

Every word has a target and a context embedding

Skip-gram learns separate parameter matrices W and C for these two roles.

$\theta =$

W target	dimensions 1 … d			
aardvark	·	·	…	·
apricot	0.31	−0.08	…	0.74
…	…			
zebra	·	·	…	·

C context	dimensions 1 … d			
aardvark	·	·	…	·
jam	−0.12	0.57	…	0.23
…	…			
zebra	·	·	…	·

Target lookup in W

$\mathbf{w} = W[\text{apricot}]$

apricot used as the target

Context lookup in C

$\mathbf{c} = C[\text{jam}]$

jam used as context or noise

Together: $2|V|$ learned vectors, each with d dimensions.

J&M Fig. 5.6 · redrawn

How can *w* and *c* express whether a target–context pair belongs together?

Dot products score target–context compatibility

A larger score supports the hypothesis that the pair occurs together in real text.

$$\mathsf{score}(\mathbf{w}, \mathbf{c}) = \mathbf{c} \cdot \mathbf{w} = \sum_j c_j w_j$$

Multiply corresponding dimensions, then add the contributions.

Illustrative 3-D values

(apricot, jam)

$\mathbf{w} = [0.60,\ 0.20,\ 0.70]$

$\mathbf{c} = [0.50,\ 0.30,\ 0.60]$

$0.30 + 0.06 + 0.42$

$\mathbf{c} \cdot \mathbf{w} = 0.78$

aligned contributions → higher compatibility

Dot products score target–context compatibility

A larger score supports the hypothesis that the pair occurs together in real text.

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aligned contributions → higher compatibility

(apricot, aardvark)

$\mathbf{w} = [0.60,\ 0.20,\ 0.70]$

$\mathbf{c} = [-0.40,\ 0.50,\ -0.30]$

$-0.24 + 0.10 - 0.21$

$\mathbf{c} \cdot \mathbf{w} = -0.35$

opposing contributions → lower compatibility

Dot products score target–context compatibility

A larger score supports the hypothesis that the pair occurs together in real text.

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$-0.24 + 0.10 - 0.21$

$\mathbf{c} \cdot \mathbf{w} = -0.35$

opposing contributions → lower compatibility

Recall: cosine is a normalized dot product. Skip-gram uses the raw dot product, so both direction and vector magnitude affect the score.

J&M §5.5.1 · Eq. 5.13

Dot products score target–context compatibility

A larger score supports the hypothesis that the pair occurs together in real text.

$$\text{score}(\mathbf{w}, \mathbf{c}) = \mathbf{c} \cdot \mathbf{w} = \sum_j c_j w_j$$

Multiply corresponding dimensions, then add the contributions.

Illustrative 3-D values

(apricot, jam)

$\mathbf{w} = [0.60,\ 0.20,\ 0.70]$

$\mathbf{c} = [0.50,\ 0.30,\ 0.60]$

$0.30 + 0.06 + 0.42$

$\mathbf{c} \cdot \mathbf{w} = 0.78$

aligned contributions → higher compatibility

(apricot, aardvark)

$\mathbf{w} = [0.60,\ 0.20,\ 0.70]$

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$\mathbf{c} \cdot \mathbf{w} = -0.35$

opposing contributions → lower compatibility

Recall: cosine is a normalized dot product. Skip-gram uses the raw dot product, so both direction and vector magnitude affect the score.

J&M §5.5.1 · Eq. 5.13

The score ranges from $-\infty$ to ∞. How can it become a probability?

The sigmoid converts compatibility into probability

The dot products from the preceding slide now become probabilities of positive pairs.

Compatibility score

$$s = \mathbf{c} \cdot \mathbf{w}$$

$$\sigma(s) = \frac{1}{1 + \exp(-s)}$$

$$P(+ \mid w, c) = \sigma(\mathbf{c} \cdot \mathbf{w})$$

Carry the same numerical scores through the transformation

apricot with jam

$s = 0.78$

$\sigma(s) = 0.686$

decision midpoint

$s = 0.00$

$\sigma(s) = 0.500$

apricot with matrix

$s = -0.35$

$\sigma(s) = 0.413$

The sigmoid converts compatibility into probability

The dot products from the preceding slide now become probabilities of positive pairs.

Compatibility score

$s = \mathbf{c} \cdot \mathbf{w}$ → $\sigma(s) = \dfrac{1}{1 + \exp(-s)}$ → $P(+ \mid w, c) = \sigma(\mathbf{c} \cdot \mathbf{w})$

Carry the same numerical scores through the transformation

apricot with jam	decision midpoint	apricot with matrix
$s = 0.78$	$s = 0.00$	$s = -0.35$
$\sigma(s) = 0.686$	$\sigma(s) = 0.500$	$\sigma(s) = 0.413$

The sigmoid changes the scale, but preserves the ordering of the scores.

J&M §5.5.1 · Eqs. 5.14–5.16

One target has several context words. How are their probabilities combined?

A context window contributes several probabilities

Conditional independence lets skip-gram multiply the pair probabilities for one target.

Shared target

$w = \text{apricot}$

tablespoon

$p_1 = \sigma(\mathbf{c}_1 \cdot \mathbf{w})$

of

$p_2 = \sigma(\mathbf{c}_2 \cdot \mathbf{w})$

jam

$p_3 = \sigma(\mathbf{c}_3 \cdot \mathbf{w})$

a

$p_4 = \sigma(\mathbf{c}_4 \cdot \mathbf{w})$

A context window contributes several probabilities

Conditional independence lets skip-gram multiply the pair probabilities for one target.

Shared target

$w =$ apricot

tablespoon

$p_1 = \sigma(\mathbf{c}_1 \cdot \mathbf{w})$

of

$p_2 = \sigma(\mathbf{c}_2 \cdot \mathbf{w})$

jam

$p_3 = \sigma(\mathbf{c}_3 \cdot \mathbf{w})$

a

$p_4 = \sigma(\mathbf{c}_4 \cdot \mathbf{w})$

Assumption: after the target is known, the context outcomes are treated as independent.

A context window contributes several probabilities

Conditional independence lets skip-gram multiply the pair probabilities for one target.

Shared target

$w = \text{apricot}$

tablespoon

$p_1 = \sigma(\mathbf{c}_1 \cdot \mathbf{w})$

of

$p_2 = \sigma(\mathbf{c}_2 \cdot \mathbf{w})$

jam

$p_3 = \sigma(\mathbf{c}_3 \cdot \mathbf{w})$

a

$p_4 = \sigma(\mathbf{c}_4 \cdot \mathbf{w})$

Assumption: after the target is known, the context outcomes are treated as independent.

$$P(+ \mid w, c_{1:4}) = \prod_{i=1}^{4} \sigma(\mathbf{c}_i \cdot \mathbf{w}) = p_1 p_2 p_3 p_4$$

J&M Eqs. 5.17–5.18

A context window contributes several probabilities

Conditional independence lets skip-gram multiply the pair probabilities for one target.

Shared target

$w = \text{apricot}$

tablespoon

$p_1 = \sigma(\mathbf{c}_1 \cdot \mathbf{w})$

of

$p_2 = \sigma(\mathbf{c}_2 \cdot \mathbf{w})$

jam

$p_3 = \sigma(\mathbf{c}_3 \cdot \mathbf{w})$

a

$p_4 = \sigma(\mathbf{c}_4 \cdot \mathbf{w})$

Assumption: after the target is known, the context outcomes are treated as independent.

$$P(+ \mid w, c_{1:4}) = \prod_{i=1}^{4} \sigma(\mathbf{c}_i \cdot \mathbf{w}) = p_1 p_2 p_3 p_4$$

J&M Eqs. 5.17–5.18

Logarithms turn the window product into a sum

The same objective becomes additive: every context word contributes one log-probability.

Window probability

$$P = p_1 p_2 p_3 p_4$$

take log

Log window probability

$$\log P = \log p_1 + \log p_2 + \log p_3 + \log p_4$$

The transformation changes the calculation, not the preferred parameters

Because log is increasing, maximizing P also maximizes log P.

J&M Eq. 5.18

How do these context pairs become positive and negative training examples?

A context window supplies positive and negative training pairs

Observed neighbours are positive; sampled vocabulary words provide contrasting negatives.

One ±2 context window around the target

tablespoon | of | **apricot** | jam | a

Observed neighbours → positive pairs

```
(apricot, tablespoon)    +
(apricot, of)    +
(apricot, jam)    +
(apricot, a)    +
```

A context window supplies positive and negative training pairs

Observed neighbours are positive; sampled vocabulary words provide contrasting negatives.

One ±2 context window around the target

tablespoon | of | apricot | jam | a

Observed neighbours → positive pairs

(apricot, tablespoon) +

(apricot, of) +

(apricot, jam) +

(apricot, a) +

Each positive pair receives k noise contrasts

(apricot, jam) +

sample k = 2

(apricot, matrix) −

(apricot, Tolstoy) −

A context window supplies positive and negative training pairs

Observed neighbours are positive; sampled vocabulary words provide contrasting negatives.

One ±2 context window around the target

tablespoon | of | **apricot** | jam | a

Observed neighbours → positive pairs

- `(apricot, tablespoon)` +
- `(apricot, of)` +
- `(apricot, jam)` +
- `(apricot, a)` +

Each positive pair receives k noise contrasts

`(apricot, jam)` +

sample k = 2

- `(apricot, matrix)` −
- `(apricot, Tolstoy)` −

Four observed neighbours give four positive pairs and, with k = 2, eight noise pairs.

J&M §5.5.2

A context window supplies positive and negative training pairs

Observed neighbours are positive; sampled vocabulary words provide contrasting negatives.

One ±2 context window around the target

tablespoon	of	**apricot**	jam	a

Observed neighbours → positive pairs

- `(apricot, tablespoon)` +
- `(apricot, of)` +
- `(apricot, jam)` +
- `(apricot, a)` +

Each positive pair receives k noise contrasts

`(apricot, jam)` +

sample k = 2

- `(apricot, matrix)` −
- `(apricot, Tolstoy)` −

Four observed neighbours give four positive pairs and, with k = 2, eight noise pairs.

J&M §5.5.2

How should SGNS choose the noise words?

The 0.75 power flattens the noise distribution

SGNS transforms word counts and then renormalizes them into sampling probabilities.

Flatten first; normalize second

The 0.75 power flattens the noise distribution

SGNS transforms word counts and then renormalizes them into sampling probabilities.

Flatten first; normalize second

1 Start from counts

$$\text{count}(w)$$

Common words begin with much larger values.

The 0.75 power flattens the noise distribution

SGNS transforms word counts and then renormalizes them into sampling probabilities.

Flatten first; normalize second

1 Start from counts

$$\text{count}(w)$$

Common words begin with much larger values.

2 Apply the 0.75 power

$$\text{count}(w)^{0.75}$$

Large count ratios shrink while ordering is preserved.

The 0.75 power flattens the noise distribution

SGNS transforms word counts and then renormalizes them into sampling probabilities.

Flatten first; normalize second

1 Start from counts

$\text{count}(w)$

Common words begin with much larger values.

2 Apply the 0.75 power

$\text{count}(w)^{0.75}$

Large count ratios shrink while ordering is preserved.

A $100{:}1$ count ratio becomes $100^{0.75}{:}1^{0.75} \approx 31.6{:}1$ before normalization.

Frequent words remain more likely, but rare words receive more sampling mass.

J&M §5.5.2 · Eqs. 5.19–5.20

The 0.75 power flattens the noise distribution

SGNS transforms word counts and then renormalizes them into sampling probabilities.

Flatten first; normalize second

1 Start from counts

$$\text{count}(w)$$

Common words begin with much larger values.

2 Apply the 0.75 power

$$\text{count}(w)^{0.75}$$

Large count ratios shrink while ordering is preserved.

3 Renormalize

$$P_{0.75}(w) = \frac{\text{count}(w)^{0.75}}{\sum_{w'} \text{count}(w')^{0.75}}$$

The transformed values now sum to one.

A $100{:}1$ count ratio becomes $100^{0.75}{:}1^{0.75} \approx 31.6{:}1$ before normalization.

Frequent words remain more likely, but rare words receive more sampling mass.

J&M §5.5.2 · Eqs. 5.19–5.20

The 0.75 power flattens the noise distribution

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Frequent words remain more likely, but rare words receive more sampling mass.

J&M §5.5.2 · Eqs. 5.19–5.20

Once the positive and negative pairs are assembled, what objective trains W and C?

SGNS maximizes correct-label probability

One case contains one observed context word and k independently sampled noise words.

One sampled training case

target w	positive context c_{pos}	noise context $c_{\text{neg},1}$	noise context $c_{\text{neg},2}$
apricot	jam	matrix	Tolstoy

SGNS maximizes correct-label probability

One case contains one observed context word and k independently sampled noise words.

One sampled training case

target w	positive context c_{pos}	noise context $c_{\text{neg},1}$	noise context $c_{\text{neg},2}$
`apricot`	`jam`	`matrix`	`Tolstoy`

Observed pair should receive +

$$P(+ \mid w, c_{\text{pos}}) = \sigma(\mathbf{c}_{\text{pos}} \cdot \mathbf{w})$$

Each noise pair should receive −

$$P(- \mid w, c_{\text{neg},i}) = \sigma(-\mathbf{c}_{\text{neg},i} \cdot \mathbf{w})$$

SGNS maximizes correct-label probability

One case contains one observed context word and k independently sampled noise words.

One sampled training case

target w	positive context c_{pos}	noise context $c_{\text{neg},1}$	noise context $c_{\text{neg},2}$
apricot	jam	matrix	Tolstoy

Observed pair should receive +

$$P(+ \mid w, c_{\text{pos}}) = \sigma(\mathbf{c}_{\text{pos}} \cdot \mathbf{w})$$

Each noise pair should receive −

$$P(- \mid w, c_{\text{neg},i}) = \sigma(-\mathbf{c}_{\text{neg},i} \cdot \mathbf{w})$$

Correct-label likelihood

$$\mathcal{L} = \sigma(\mathbf{c}_{\text{pos}} \cdot \mathbf{w}) \prod_{i=1}^{k} \sigma(-\mathbf{c}_{\text{neg},i} \cdot \mathbf{w})$$

J&M §5.5.2 · Eq. 5.21

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

Start with the score for one sampled noise pair

1 Compatibility score

$$s_{\text{neg}} = c_{\text{neg}} \cdot w$$

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

Start with the score for one sampled noise pair

1 Compatibility score

$$s_{\text{neg}} = c_{\text{neg}} \cdot w$$

2 Positive-class probability

$$P(+ \mid w, c_{\text{neg}}) = \sigma(s_{\text{neg}})$$

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

Start with the score for one sampled noise pair

1 Compatibility score

$$s_{\text{neg}} = c_{\text{neg}} \cdot w$$

2 Positive-class probability

$$P(+ \mid w, c_{\text{neg}}) = \sigma(s_{\text{neg}})$$

3 A negative label means "not positive"

$$P(- \mid w, c_{\text{neg}}) = 1 - P(+ \mid w, c_{\text{neg}}) = 1 - \sigma(s_{\text{neg}})$$

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

Start with the score for one sampled noise pair

1 Compatibility score

$$s_{\text{neg}} = c_{\text{neg}} \cdot w$$

2 Positive-class probability

$$P(+ \mid w, c_{\text{neg}}) = \sigma(s_{\text{neg}})$$

3 A negative label means "not positive"

$$P(- \mid w, c_{\text{neg}}) = 1 - P(+ \mid w, c_{\text{neg}}) = 1 - \sigma(s_{\text{neg}})$$

Sigmoid complement identity

$$1 - \sigma(s) = \sigma(-s)$$

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

Start with the score for one sampled noise pair

1 Compatibility score

$$s_{\text{neg}} = c_{\text{neg}} \cdot w$$

2 Positive-class probability

$$P(+ \mid w, c_{\text{neg}}) = \sigma(s_{\text{neg}})$$

3 A negative label means “not positive”

$$P(- \mid w, c_{\text{neg}}) = 1 - P(+ \mid w, c_{\text{neg}}) = 1 - \sigma(s_{\text{neg}})$$

Sigmoid complement identity

$$1 - \sigma(s) = \sigma(-s)$$

Therefore

$$P(- \mid w, c_{\text{neg}}) = \sigma(-s_{\text{neg}}) = \sigma\big(-(c_{\text{neg}} \cdot w)\big)$$

The negative label reverses the compatibility score

For a noise pair, the correct-class probability is the complement of the positive-class probability.

Start with the score for one sampled noise pair

1 Compatibility score

$$s_{\text{neg}} = c_{\text{neg}} \cdot w$$

2 Positive-class probability

$$P(+ \mid w, c_{\text{neg}}) = \sigma(s_{\text{neg}})$$

3 A negative label means "not positive"

$$P(- \mid w, c_{\text{neg}}) = 1 - P(+ \mid w, c_{\text{neg}}) = 1 - \sigma(s_{\text{neg}})$$

Sigmoid complement identity

$$1 - \sigma(s) = \sigma(-s)$$

Therefore

$$P(- \mid w, c_{\text{neg}}) = \sigma(-s_{\text{neg}}) = \sigma\big(-(c_{\text{neg}} \cdot w)\big)$$

The outer minus negates the scalar score—not a separately stored context vector.

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Numerical check

Suppose the noise pair already has a low compatibility score:

$$s_{\text{neg}} = -3$$

$$P(+) = \sigma(-3) = 0.047$$

$$P(-) = 1 - 0.047 = 0.953$$

$$P(-) = \sigma(3) = 0.953$$

A negative score gives a high probability to the negative label.

$$P(\text{correct}) = \sigma(ys)$$

y = +1 for observed; y = −1 for noise

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The negative log gives the SGNS loss

The positive term and all k negative terms become additive contributions.

$$\mathcal{L} = \sigma(\mathbf{c}_{\text{pos}} \cdot \mathbf{w}) \prod_{i=1}^{k} \sigma(-\mathbf{c}_{\text{neg},i} \cdot \mathbf{w})$$

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SGNS loss

$$L = -\left[\log \sigma(\mathbf{c}_{\text{pos}} \cdot \mathbf{w}) + \sum_{i=1}^{k} \log \sigma(-\mathbf{c}_{\text{neg},i} \cdot \mathbf{w})\right]$$

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Positive term

Minimizing L increases $\mathbf{c}_{\text{pos}} \cdot \mathbf{w}$.

Negative terms

Minimizing L decreases every $\mathbf{c}_{\text{neg},i} \cdot \mathbf{w}$.

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J&M §5.5.2 · Eq. 5.21

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J&M §5.5.2 · Eq. 5.21

What does one complete forward pass look like with actual numbers?

Running text supplies the positive examples

A context window converts one sentence into labelled target–context pairs.

Sentence

I spread **apricot** jam on warm toast.

window $= 1$ word on each side

observed near apricot

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The target word is paired with every word inside its window

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`(apricot, spread)` $y = 1$

observed immediately before

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(apricot, spread)	$y = 1$	observed immediately before
(apricot, jam)	$y = 1$	observed immediately after

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window $= 1$ word on each side

observed near apricot

The target word is paired with every word inside its window

(apricot, spread)	$y = 1$	observed immediately before
(apricot, jam)	$y = 1$	observed immediately after

These labels come from the text itself: no human annotator is required.

J&M §5.5.2

Negative sampling creates the contrasting examples

Keep one observed pair, then sample words that did not occupy that context position.

One small training batch		*meaning of the label*
`(apricot, jam)`	$y = 1$	jam occurred beside apricot
`(apricot, aardvark)`	$y = 0$	sampled noise word
`(apricot, tractor)`	$y = 0$	sampled noise word
`(apricot, ocean)`	$y = 0$	sampled noise word

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One small training batch — *meaning of the label*

Pair	Label	Meaning of the label
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“Negative” means not observed in this context window. It does not mean the words can never be related.

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“Negative” means not observed in this context window. It does not mean the words can never be related.

J&M §5.5.2

Each labelled pair selects two embedding rows

The target comes from W; every candidate context word comes from C.

Target lookup

$W[\text{apricot}]$

$$\mathbf{w} = [0.60,\ 0.20,\ 0.70]$$

the same target row is reused

Context lookups

$C[\text{jam}]$ $[0.50,\ 0.30,\ 0.60]$

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$C[\text{jam}]$ $[0.50,\ 0.30,\ 0.60]$

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$C[\text{ocean}]$	$[0.10,\ -0.50,\ -0.40]$

Four pairs → one W row and four C rows. The word apricot is not re-embedded four times.

J&M §5.5.2

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$C[\text{aardvark}]$ $[-0.40,\ 0.50,\ -0.30]$

$C[\text{tractor}]$ $[-0.50,\ -0.20,\ 0.10]$

$C[\text{ocean}]$ $[0.10,\ -0.50,\ -0.40]$

Dot products score the four candidate pairs

For each pair, multiply matching dimensions and add the three contributions.

$$\text{score}(\mathbf{w}, \mathbf{c}) = \mathbf{c} \cdot \mathbf{w}$$

`jam`	$0.50(0.60) + 0.30(0.20) + 0.60(0.70) = 0.30 + 0.06 + 0.42$	$\text{score} = 0.78$
`aardvark`	$-0.40(0.60) + 0.50(0.20) - 0.30(0.70) = -0.24 + 0.10 - 0.21$	$\text{score} = -0.35$
`tractor`	$-0.50(0.60) - 0.20(0.20) + 0.10(0.70) = -0.30 - 0.04 + 0.07$	$\text{score} = -0.27$
`ocean`	$0.10(0.60) - 0.50(0.20) - 0.40(0.70) = 0.06 - 0.10 - 0.28$	$\text{score} = -0.32$

$W[\text{apricot}]$

$\mathbf{w} = [0.60,\ 0.20,\ 0.70]$

the same target row is reused

$C[\text{jam}]$	$[0.50,\ 0.30,\ 0.60]$
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$C[\text{ocean}]$	$[0.10,\ -0.50,\ -0.40]$

J&M §5.5.2

The sigmoid turns each score into a pair probability

Apply the same nonlinear transformation independently to all four scores.

$$p(y=1 \mid w, c) = \sigma(\mathbf{c} \cdot \mathbf{w}) \quad \text{where} \quad \sigma(s) = \frac{1}{1+\exp(-s)}$$

pair	score s	σ(s)	label y	reading
(apricot, jam)	0.78	0.686	1	*higher is better*
(apricot, aardvark)	−0.35	0.413	0	*lower is better*
(apricot, tractor)	−0.27	0.433	0	*lower is better*
(apricot, ocean)	−0.32	0.421	0	*lower is better*

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(apricot, ocean)	−0.32	0.421	0	*lower is better*

The sigmoid changes the scale but preserves the ordering of the scores.

J&M §5.5.2

The forward pass ends with one combined SGNS loss

Reward the observed pair for high probability and every noise pair for low probability.

Observed pair

$$-\log \sigma(\mathbf{c}_{\text{jam}} \cdot \mathbf{w})$$

$$= -\log(0.685680)$$

$$= 0.377344$$

small when the positive probability is high

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$$=0.377344$$

small when the positive probability is high

Three noise pairs

`aardvark` $-\log(1-\sigma(-0.35))=0.533382$

`tractor` $-\log(1-\sigma(-0.27))=0.567232$

`ocean` $-\log(1-\sigma(-0.32))=0.545893$

noise subtotal = 1.646507

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$$-\log \sigma(\mathbf{c}_{\text{jam}} \cdot \mathbf{w})$$

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noise subtotal = 1.646507

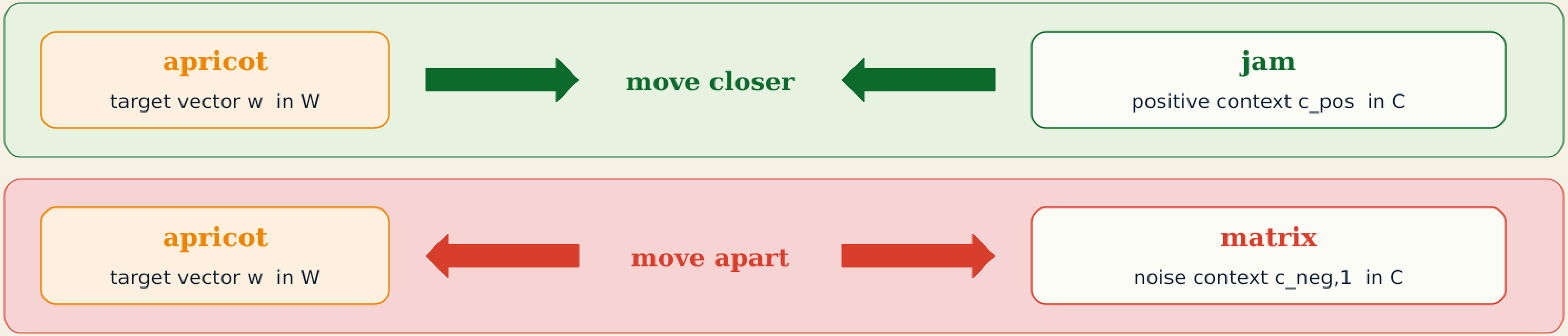
L = 0.377344 + 1.646507 = 2.023851 ≈ 2.024

Forward pass complete: no parameter has been changed yet.

J&M §5.5.2 · Eq. 5.20

SGD pulls neighbours closer and repels noise

Figure 5.7 visualizes how one update changes the sampled target and context vectors.



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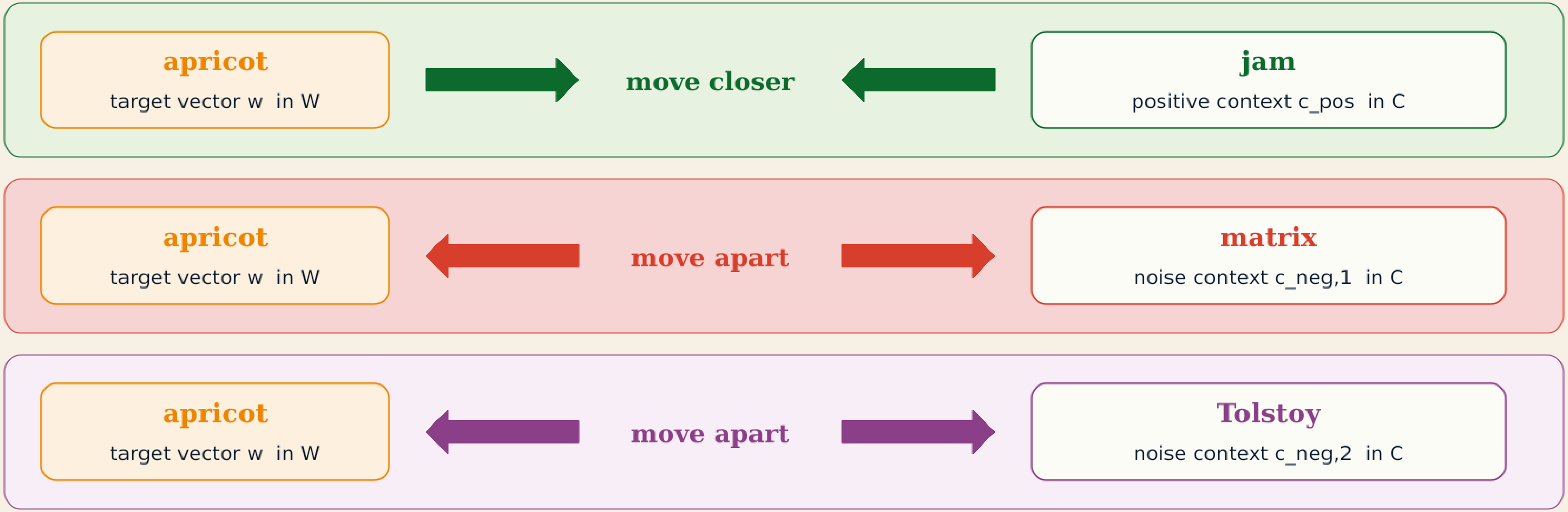

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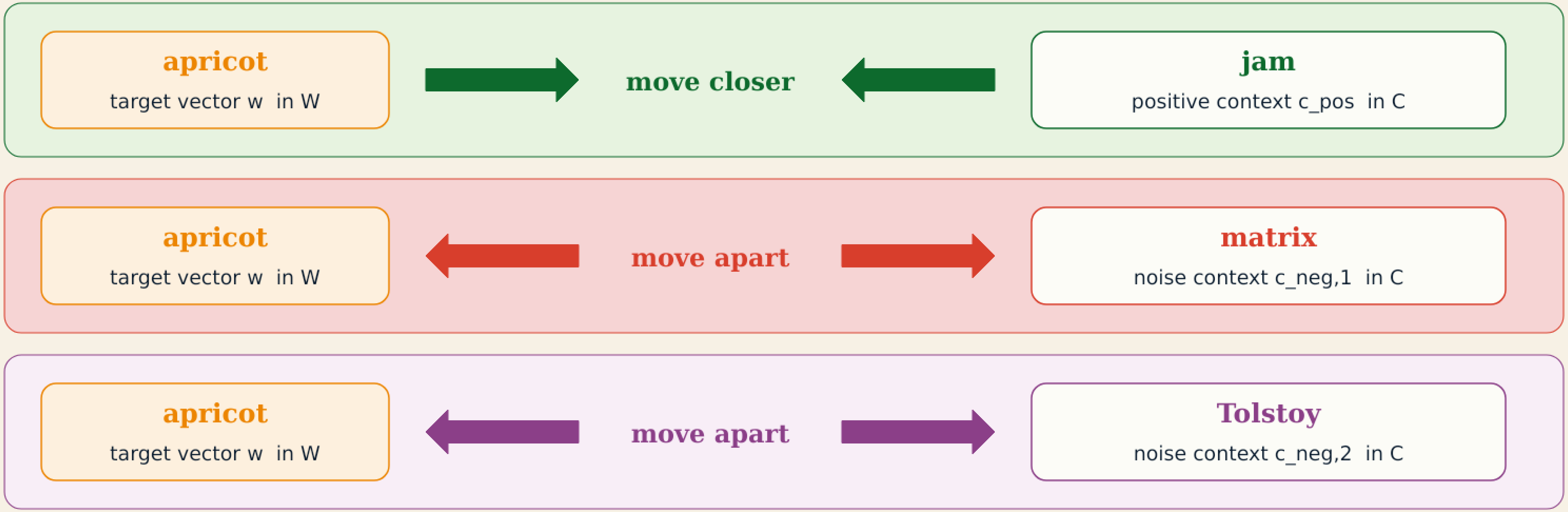

One target update combines the positive pull and both negative pushes.

J&M Fig. 5.7 · redrawn

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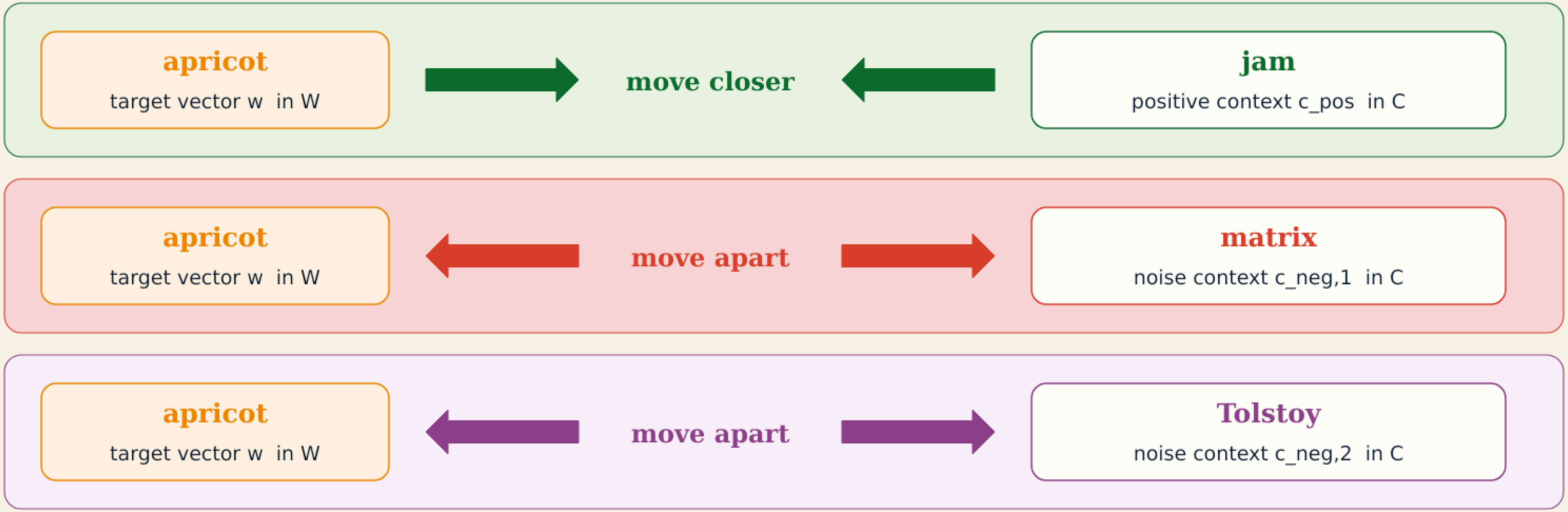

One target update combines the positive pull and both negative pushes.

J&M Fig. 5.7 · redrawn

The geometry is intuitive. What scalar coefficients determine each movement size?